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Simultaneous Equations

Two linear equations: eliminate. One linear and one quadratic: make one letter the subject and substitute. Then ERCS and solve.

THE TOPIC ON ONE MAP

Simultaneous Equations

01 Linear: elimination

match a coefficient

a constant k

02 Linear and quadratic: substitute

x^2 , xy

show $x^2 - 2x - 35 = 0$

03 One solution: the discriminant

tangent

two / no solutions

04 Models and harder forms

$H^3 = pt^2 + q$

exact distance PQ

01 Linear: Elimination

Calculator first — you know the answer. Then show it: match one coefficient, add or subtract, back-substitute.

WHEN YOU SEE → YOU DO

“use elimination”

→ multiply to match a coefficient

signs of the matched terms are different

→ ADD

signs are the same

→ SUBTRACT

a constant k in both equations

→ solve as normal — the answer contains k

WORKED MODEL • ELIMINATION

$$7x + 4y = 17, \quad 3x - 2y = 11$$

1 $7x + 4y = 17$ and $6x - 4y = 22$

second $\times 2$

2 $13x = 39 \Rightarrow x = 3$

add

ANSWER $x = 3, y = -1$

IDEA 01

Practise: Linear: Elimination

My questions first, then selected practice.

Q1 **DR ESLAM** Add or subtract

P1 · TOPIC 3 3 marks

Solve $x + 2y = 7$, $3x - y = 7$.

SELECTED PRACTICE

Q2 **PRACTICE** Practice · Easy

P1 · TOPIC 3 2 marks

1 Solve the simultaneous equations

$$x + y = 8$$

$$x - y = 4$$

(2 marks)

Q3**PRACTICE**

Practice · Medium

P1 · TOPIC 3 4 marks

- 1 Use elimination to solve the simultaneous equations

$$7x + 4y = 17$$

$$3x - 2y = 11$$

(4 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 3**5 marks****6 (a)**

$$4x - ky = 13$$

$$3x + 2ky = 7$$

are simultaneous equations, where k is a constant.

Show that $x = 3$.

(3 marks)

(b) Find an expression for y in terms of the constant k .

(1 mark)

(c) Given that $y = 3$, find the value of k .

(1 mark)

02 Linear and Quadratic: Substitute

Rearrange the linear one, substitute into the quadratic, ERCS, solve, then back into the **linear** one for the pairs.

WHEN YOU SEE → YOU DO

one equation has x^2, y^2 or xy

→ substitution — never elimination

“by eliminating y , show that $x^2 - 2x - 35 = 0$ ”

→ make y the subject of the linear, substitute

“hence solve”

→ solve the quadratic, then find each y

“in the form $a \pm b\sqrt{3}$ ”

→ quadratic formula, exact

“state which x goes with which y ”

→ write the answers as pairs

1

Rearrange the linear

Choose the letter with coefficient 1 to avoid fractions.

2

Substitute and ERCS

Expand, rearrange to $= 0$, collect.

3

Solve and pair

Two x values → two y values from the **linear** equation.

WORKED MODEL • JANUARY 2026

$$y - x = 4, \quad 2x^2 - xy = 8$$

1 $2x^2 - x(x + 4) = 8 \Rightarrow x^2 - 4x - 8 = 0$

substitute $y = x + 4$

2 $x = 2 \pm 2\sqrt{3}$

formula, exact

ANSWER $(2 + 2\sqrt{3}, 6 + 2\sqrt{3}), (2 - 2\sqrt{3}, 6 - 2\sqrt{3})$

IDEA 02

Practise: Linear and Quadratic: Substitute

My questions first, then selected practice, then real past papers.

Q5 DR ESLAM Substitute

P1 · TOPIC 3 4 marks

Solve $y = x + 1$, $x^2 + y^2 = 13$.

SELECTED PRACTICE**Q6** PRACTICE Practice · Easy

P1 · TOPIC 3 4 marks

6 Solve the simultaneous equations

$$y = 2x - 1$$

$$x^2 + y^2 - 2 = 0$$

(4 marks)

Q7**PRACTICE**

Practice · Medium

P1 · TOPIC 3**7 marks**

5 (a) By eliminating y from the equations

$$x^2 - 8y = -40$$

$$3x + 2y = 4$$

show that $x^2 + 12x + 24 = 0$.

(2 marks)

(b) Hence solve the simultaneous equations

$$x^2 - 8y = -40$$

$$3x + 2y = 4$$

giving x and y in the form $a \pm b\sqrt{3}$, where a and b are integers.

(5 marks)

Q8**PRACTICE**

Practice · Hard

P1 · TOPIC 3**7 marks**

5 (a) By eliminating y from the equations

$$15x^2 - 4y^2 = 12$$

$$4x + 2y = 1$$

show that $x^2 - 8x + 13 = 0$.

(2 marks)

(b) Hence solve the simultaneous equations

$$15x^2 - 4y^2 = 12$$

$$4x + 2y = 1$$

giving x and y in the form $a \pm b\sqrt{3}$, where a and b are rational numbers.

(5 marks)

PAST PAPERS

Q9

PAST PAPER

Pearson IAL WMA11 · January 2026 · Q2

P1 · TOPIC 3

6 marks

2.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(a) Given

$$y - x = 4$$

and

$$2x^2 - xy = 8$$

show that

$$x^2 - 4x - 8 = 0$$

(2)

(b) Hence solve the simultaneous equations

$$y - x = 4$$

$$2x^2 - xy = 8$$

writing the answers in the form $p + q\sqrt{3}$, where p and q are integers to be found.

(4)

(Total for Question 2 is 6 marks)

03 One Solution: the Discriminant

“The line touches the curve”, “exactly one solution” — the quadratic you get after substituting has $b^2 - 4ac = 0$.

WHEN YOU SEE → YOU DO

“exactly one solution”, “the line is a tangent”

$$\longrightarrow b^2 - 4ac = 0$$

“two solutions / no solutions”

$$\longrightarrow > 0 \text{ / } < 0$$

“find the solution for that k ”

$$\longrightarrow \text{repeated root } x = -\frac{b}{2a}$$

WORKED MODEL • ONE SOLUTION

$$x^2 - 2y = 10, \quad 2x - y = k$$

$$1 \quad x^2 - 2(2x - k) = 10 \Rightarrow x^2 - 4x + 2(k - 5) = 0$$

substitute

$$2 \quad 16 - 8(k - 5) = 0 \Rightarrow k = 7$$

$$b^2 - 4ac = 0$$

ANSWER $k = 7, \quad x = 2, \quad y = -3$

IDEA 03

Practise: One Solution: the Discriminant

My questions first, then selected practice.

Q10 DR ESLAM Tangent line

P1 · TOPIC 3 3 marks

Find k such that the line $y = 2x + k$ touches the curve $y = x^2 + 4$.

$k =$ _____

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Medium

P1 · TOPIC 3 8 marks

7 (a)

$$x^2 - 2y = 10$$

$$2x - y = k$$

are simultaneous equations, where k is a constant.

By eliminating y from the equations show that $x^2 - 4x + 2(k - 5) = 0$.

(3 marks)

- (b) By considering the discriminant of $x^2 - 4x + 2(k - 5) = 0$ find the value of k for which the simultaneous equations have only one solution.

(2 marks)

- (c) Find the solution to the simultaneous equations for the value of k that you found in part (b).

(3 marks)

Q12**PRACTICE**

Practice · Hard

P1 · TOPIC 3 8 marks**7 (a)**

$$x^2 - y + 3x = k$$

$$20x - 4y = -5$$

are simultaneous equations, where k is a constant.

Given that the simultaneous equations have exactly one solution, find the value of the constant k .

(5 marks)

- (b)** Find the solution to the simultaneous equations for the value of k that you found in part (a).

(3 marks)

Q13**PRACTICE**

Practice · Very Hard

P1 · TOPIC 3 9 marks**7 (a)**

$$x^2 + 2y^2 = 25$$

$$x - y = k$$

are simultaneous equations, where k is a constant.

Find the respective sets of values for k for which the simultaneous equations have one, two, and no solutions.

(6 marks)

- (b)** Given that the simultaneous equations have exactly one solution, find all possible pairs (x, y) that might correspond to that solution. Give all your values for x and y in the form $a\sqrt{6}$, where a is a rational number.

(3 marks)

04 Models and Harder Forms

A model $H^3 = pt^2 + q$ with two data points is two linear equations in p and q . Quartics $2x^4 - 7x^2 - 15 = 0$ are hidden quadratics.

WHEN YOU SEE → YOU DO

“given $H^3 = pt^2 + q$... when $t = 3, H = 2$ ” → substitute both points → solve for p, q

“ $2x^4 - 15x^2 - 50 = 0$ ” → let $u = x^2$

“the exact distance PQ ” → $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

a curve with xy → make y the subject: $y = \frac{\dots}{x}$

WORKED MODEL • JANUARY 2021

$H^3 = pt^2 + q$; $t = 3, H = 2$ and $t = 5, H = 2.4$.

1 $8 = 9p + q, 13.824 = 25p + q$

two data points

2 $p = 0.364, q = 4.72$

subtract

3 $125 = 0.364 T^2 + 4.72$

height 5 m

ANSWER $T = 18.2$ years

IDEA 04

Practise: Models and Harder Forms

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q14**

PRACTICE

Practice • Medium

P1 • TOPIC 3

6 marks

- 8 You are asked to advise a client on which parcel delivery service to use to deliver parcels of differing sizes. Linear Deliveries Inc. charges a flat rate of £2.25 per parcel, plus 40p times the mass of the parcel in kilograms. Square Deal Delivery Solutions charges a flat rate of £4 per parcel, plus 1p times the *square* of the parcel's mass in kilograms. Under what circumstances would you advise your client to use each of the two delivery services? Be sure to show clear mathematical justifications for your answer.

(6 marks)

Q15**PRACTICE**

Practice · Hard

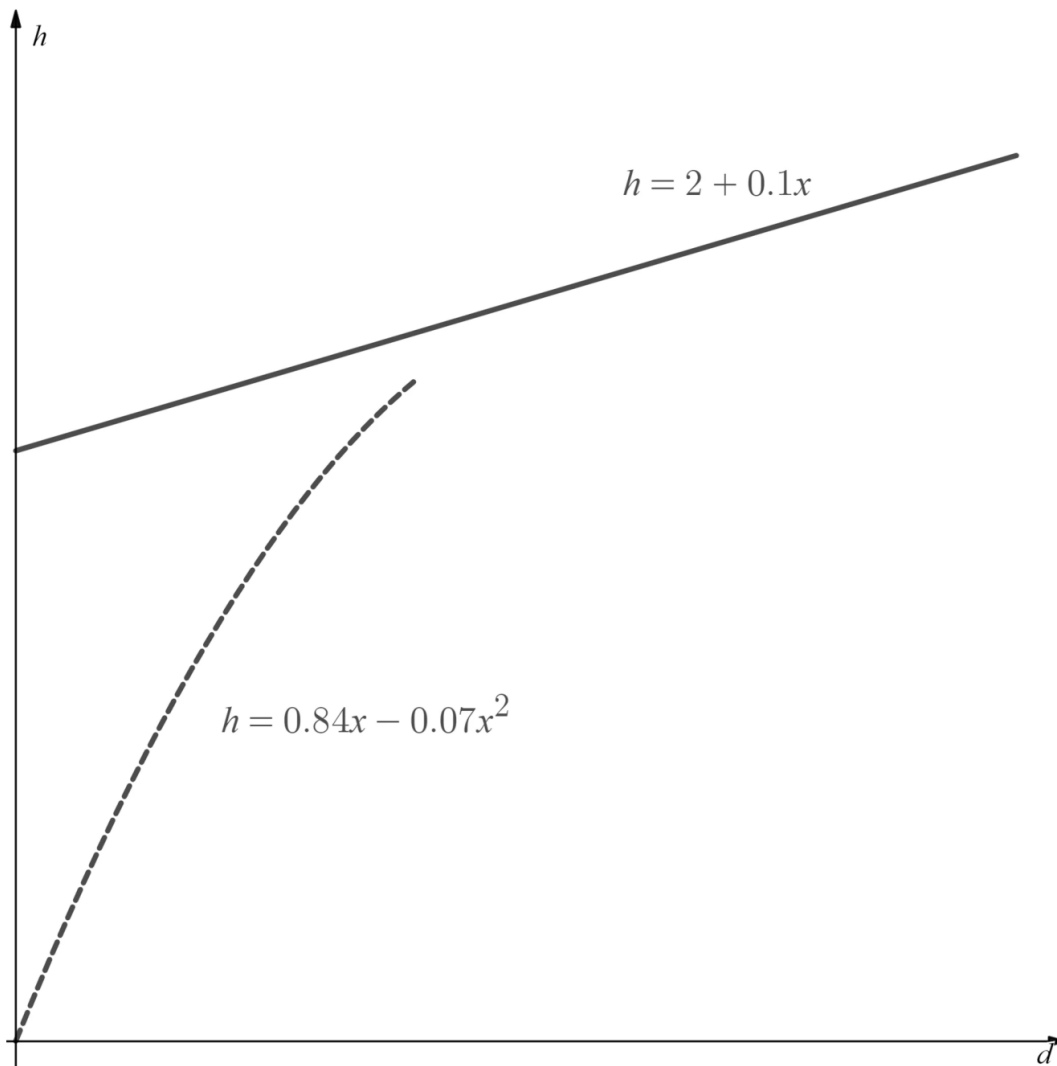
P1 · TOPIC 3**5 marks**

- 8 A firework is launched inside a large shed with a sloping roof. In relation to the horizontal distance from the point it was launched, the height of the firework, h m, can be modelled by the quadratic equation

$$h = 0.84x - 0.07x^2$$

The sloping roof of the shed can be modelled with the equation

$$h = 2 + 0.1x$$



Determine whether, according to the model, the firework will hit the roof of the shed before escaping out the open end of the shed on the right of the diagram.

(5 marks)

PAST PAPERS

Q16 PAST PAPER Pearson IAL WMA11 • January 2021 • Q2

P1 • TOPIC 3 6 marks

2. A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the equation

$$H^3 = pt^2 + q$$

where p and q are constants,

- (a) find, to 3 significant figures where necessary, the value of p and the value of q .

(4)

Exactly T years after the tree was planted, its height was 5 m.

- (b) Find the value of T according to the model, giving your answer to one decimal place.

(2)

(Total 6 marks)

Q17 **PAST PAPER** Pearson IAL WMA11 · June 2022 · Q6**P1 · TOPIC 3** **7 marks**

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0 \quad (2)$$

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

(Total 7 marks)

Q18**PAST PAPER**

Pearson IAL WMA11 · October 2023 · Q8

P1 · TOPIC 3 7 marks

8. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0 \quad (2)$$

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ (5)

(Total for Question 8 is 7 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · January 2025 · Q3

P1 · TOPIC 3 5 marks

3. The population of a town was monitored.

Exactly 5 years after monitoring began, the population was 58 000

Exactly 10 years after monitoring began, the population was 65 000

Given that the population of the town, P thousand, t years after monitoring began can be modelled by the equation

$$P^2 = a + bt^3$$

where a and b are constants,

- (a) find the value of a and the value of b .

(3)

According to the model, exactly T years after monitoring began, the population was 85 000

Making your method clear,

- (b) find the value of T , giving your answer to one decimal place.

(Solutions relying entirely on calculator technology are not acceptable.)

(2)**(Total for Question 3 is 5 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | | |
|-----------|-------------------------|---|---|
| 01 | Linear | | Match a coefficient → different signs add, same signs subtract. |
| 02 | Linear quadratic | + | Subject from the linear → substitute → ERCS → pairs. |
| 03 | One solution | | Discriminant of the substituted quadratic = 0. |
| 04 | Models | | Two data points → two equations in p, q . $x^4 \rightarrow u = x^2$. |

BOTTOM LINE Calculator first → eliminate (linear) or substitute (quadratic) → ERCS → pair every x with its y .

Answers

Simultaneous Equations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LINEAR: ELIMINATION

- | | | |
|-----------|----------------------------|---|
| Q1 | Dr Eslam • Add or subtract | $x = 3, y = 2$ |
| Q2 | Practice • Easy | $x = 6, y = 2$ |
| Q3 | Practice • Medium | $x = 3, y = -1$ |
| Q4 | Practice • Medium | (b) $y = -\frac{1}{k}$ (c) $k = -\frac{1}{3}$ |

IDEA 02 • LINEAR AND QUADRATIC: SUBSTITUTE

- | | | |
|-----------|---------------------------------------|--|
| Q5 | Dr Eslam • Substitute | $(2, 3)$ and $(-3, -2)$ |
| Q6 | Practice • Easy | $(1, 1)$ and $(-\frac{1}{5}, -\frac{7}{5})$ |
| Q7 | Practice • Medium | (b) $x = -6 \pm 2\sqrt{3}, y = 11 \mp 3\sqrt{3}$ |
| Q8 | Practice • Hard | (b) $x = 4 \mp \sqrt{3}, y = -\frac{15}{2} \pm 2\sqrt{3}$ |
| Q9 | Pearson IAL WMA11 • January 2026 • Q2 | (b) $(2 + 2\sqrt{3}, 6 + 2\sqrt{3}), (2 - 2\sqrt{3}, 6 - 2\sqrt{3})$ |

IDEA 03 • ONE SOLUTION: THE DISCRIMINANT

- | | | |
|------------|-------------------------|---|
| Q10 | Dr Eslam • Tangent line | $x^2 - 2x + 4 - k = 0: 4 - 4(4 - k) = 0 \Rightarrow k = 3$ |
| Q11 | Practice • Medium | (b) $k = 7$ (c) $x = 2, y = -3$ |
| Q12 | Practice • Hard | (a) $k = -\frac{9}{4}$ (b) $x = 1, y = \frac{25}{4}$ |
| Q13 | Practice • Very Hard | one: $k = \pm \frac{5\sqrt{6}}{2}$; two: $ k < \frac{5\sqrt{6}}{2}$; none: $ k > \frac{5\sqrt{6}}{2}$ |

IDEA 04 • MODELS AND HARDER FORMS

- Q14** Practice • Medium equal cost at $m = 5$ and $m = 35$ kg: Square Deal is cheaper for $5 < m < 35$, Linear otherwise
- Q15** Practice • Hard $0.07x^2 - 0.74x + 2 = 0$ has no real roots ($b^2 - 4ac < 0$): it escapes without hitting the roof
(a) $p = 0.364$, $q = 4.72$ (b) $T = 18.2$
- Q16** Pearson IAL WMA11 • January 2021 • Q2
- Q17** Pearson IAL WMA11 • June 2022 • Q6
(b) $x = \sqrt{10}$, $y = 4\sqrt{10}$; $x = -\sqrt{10}$, $y = -4\sqrt{10}$
- Q18** Pearson IAL WMA11 • October 2023 • Q8
(b) $PQ = \sqrt{65}$
- Q19** Pearson IAL WMA11 • January 2025 • Q3
(a) $a = 3241$, $b = \frac{123}{125}$ (b) $T = 15.9$